

xtivreg — Instrumental variables and two-stage least squares for panel-data models

Syntax	Menu	Description
Options for RE model	Options for BE model	Options for FE model
Options for FD model	Remarks and examples	Stored results
Methods and formulas	Acknowledgment	References
Also see		

Syntax

GLS random-effects (RE) model

```
xtivreg depvar [varlist1] (varlist2 = varlistiv) [if] [in] [, re RE_options]
```

Between-effects (BE) model

```
xtivreg depvar [varlist1] (varlist2 = varlistiv) [if] [in] , be [BE_options]
```

Fixed-effects (FE) model

```
xtivreg depvar [varlist1] (varlist2 = varlistiv) [if] [in] , fe [FE_options]
```

First-differenced (FD) estimator

```
xtivreg depvar [varlist1] (varlist2 = varlistiv) [if] [in] , fd [FD_options]
```

RE_options	Description
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Model	
re	use random-effects estimator; the default
ec2sls	use Baltagi's EC2SLS random-effects estimator
nosa	use the Baltagi–Chang estimators of the variance components
regress	treat covariates as exogenous and ignore instrumental variables
SE	
vce(vcetype)	vcetype may be conventional, bootstrap, or jackknife
Reporting	
level(#)	set confidence level; default is level(95)
first	report first-stage estimates
small	report <i>t</i> and <i>F</i> statistics instead of <i>Z</i> and χ^2 statistics
theta	report θ
display_options	control column formats, row spacing, line width, display of omitted variables and base and empty cells, and factor-variable labeling
coeflegend	display legend instead of statistics

<i>BE_options</i>	Description
Model	
<code>be</code>	use between-effects estimator
<code>regress</code>	treat covariates as exogenous and ignore instrumental variables
SE	
<code>vce(<i>vcetype</i>)</code>	<i>vcetype</i> may be <code>conventional</code> , <code>bootstrap</code> , or <code>jackknife</code>
Reporting	
<code>level(#)</code>	set confidence level; default is <code>level(95)</code>
<code>first</code>	report first-stage estimates
<code>small</code>	report <i>t</i> and <i>F</i> statistics instead of <i>Z</i> and χ^2 statistics
<code>display_options</code>	control column formats, row spacing, line width, display of omitted variables and base and empty cells, and factor-variable labeling
<code>coeflegend</code>	display legend instead of statistics
<i>FE_options</i>	Description
Model	
<code>fe</code>	use fixed-effects estimator
<code>regress</code>	treat covariates as exogenous and ignore instrumental variables
SE	
<code>vce(<i>vcetype</i>)</code>	<i>vcetype</i> may be <code>conventional</code> , <code>bootstrap</code> , or <code>jackknife</code>
Reporting	
<code>level(#)</code>	set confidence level; default is <code>level(95)</code>
<code>first</code>	report first-stage estimates
<code>small</code>	report <i>t</i> and <i>F</i> statistics instead of <i>Z</i> and χ^2 statistics
<code>display_options</code>	control column formats, row spacing, line width, display of omitted variables and base and empty cells, and factor-variable labeling
<code>coeflegend</code>	display legend instead of statistics

<i>FD_options</i>	Description
Model	
<code>noconstant</code>	suppress constant term
<code>fd</code>	use first-differenced estimator
<code>regress</code>	treat covariates as exogenous and ignore instrumental variables
SE	
<code>vce(<i>vcetype</i>)</code>	<i>vcetype</i> may be <code>conventional</code> , <code>bootstrap</code> , or <code>jackknife</code>
Reporting	
<code>level(#)</code>	set confidence level; default is <code>level(95)</code>
<code>first</code>	report first-stage estimates
<code>small</code>	report <i>t</i> and <i>F</i> statistics instead of <i>Z</i> and χ^2 statistics
<code>display_options</code>	control column formats, row spacing, line width, and display of omitted variables
<code>coeflegend</code>	display legend instead of statistics

A panel variable must be specified. For `xtivreg`, `fd` a time variable must also be specified. Use `xtset`; see [XT] `xtset`.
`varlist1` and `varlisttv` may contain factor variables, except for the `fd` estimator; see [U] 11.4.3 Factor variables.
`devar`, `varlist1`, `varlist2`, and `varlisttv` may contain time-series operators; see [U] 11.4.4 Time-series varlists.
`by` and `statsby` are allowed; see [U] 11.1.10 Prefix commands.
`coeflegend` does not appear in the dialog box.
See [U] 20 Estimation and postestimation commands for more capabilities of estimation commands.

Menu

Statistics > Longitudinal/panel data > Endogenous covariates > Instrumental-variables regression (FE, RE, BE, FD)

Description

`xtivreg` offers five different estimators for fitting panel-data models in which some of the right-hand-side covariates are endogenous. These estimators are two-stage least-squares generalizations of simple panel-data estimators for exogenous variables. `xtivreg` with the `be` option uses the two-stage least-squares between estimator. `xtivreg` with the `fe` option uses the two-stage least-squares within estimator. `xtivreg` with the `re` option uses a two-stage least-squares random-effects estimator. There are two implementations: G2SLS from Balestra and Varadharajan-Krishnakumar (1987) and EC2SLS from Baltagi. The Balestra and Varadharajan-Krishnakumar G2SLS is the default because it is computationally less expensive. Baltagi’s EC2SLS can be obtained by specifying the `ec2sls` option. `xtivreg` with the `fd` option requests the two-stage least-squares first-differenced estimator.

See Baltagi (2013) for an introduction to panel-data models with endogenous covariates. For the derivation and application of the first-differenced estimator, see Anderson and Hsiao (1981).

Options for RE model

Model

re requests the G2SLS random-effects estimator. **re** is the default.

ec2sls requests Baltagi's EC2SLS random-effects estimator instead of the default Balestra and Varadharajan-Krishnakumar estimator.

nosa specifies that the Baltagi–Chang estimators of the variance components be used instead of the default adapted Swamy–Arora estimators.

regress specifies that all the covariates be treated as exogenous and that the instrument list be ignored. Specifying **regress** causes **xtivreg** to fit the requested panel-data regression model of *depvar* on *varlist*₁ and *varlist*₂, ignoring *varlist*_{iv}.

SE

vce(*vcetype*) specifies the type of standard error reported, which includes types that are derived from asymptotic theory (**conventional**) and that use bootstrap or jackknife methods (**bootstrap**, **jackknife**); see [XT] *vce_options*.

vce(**conventional**), the default, uses the conventionally derived variance estimator for generalized least-squares regression.

Reporting

level(#); see [R] *estimation options*.

first specifies that the first-stage regressions be displayed.

small specifies that *t* statistics be reported instead of *Z* statistics and that *F* statistics be reported instead of χ^2 statistics.

theta specifies that the output include the estimated value of θ used in combining the between and fixed estimators. For balanced data, this is a constant, and for unbalanced data, a summary of the values is presented in the header of the output.

display_options: **noomitted**, **vsquish**, **noemptycells**, **baselevels**, **allbaselevels**, **nofvlabel**, **fvwrap**(#), **fvwrapon**(*style*), **cformat**(%*fnt*), **pformat**(%*fnt*), **sformat**(%*fnt*), and **nolstretch**; see [R] *estimation options*.

The following option is available with **xtivreg** but is not shown in the dialog box:

coeflegend; see [R] *estimation options*.

Options for BE model

Model

be requests the between regression estimator.

regress specifies that all the covariates are to be treated as exogenous and that the instrument list is to be ignored. Specifying **regress** causes **xtivreg** to fit the requested panel-data regression model of *depvar* on *varlist*₁ and *varlist*₂, ignoring *varlist*_{iv}.

SE

`vce(vcetype)` specifies the type of standard error reported, which includes types that are derived from asymptotic theory (`conventional`) and that use bootstrap or jackknife methods (`bootstrap`, `jackknife`); see [XT] [vce_options](#).

`vce(conventional)`, the default, uses the conventionally derived variance estimator for generalized least-squares regression.

Reporting

`level(#)`; see [R] [estimation options](#).

`first` specifies that the first-stage regressions be displayed.

`small` specifies that t statistics be reported instead of Z statistics and that F statistics be reported instead of χ^2 statistics.

display_options: `noomitted`, `vsquish`, `noemptycells`, `baselevels`, `allbaselevels`, `nofvlabel`, `fvwrap(#)`, `fvwrapon(style)`, `cformat(%fmt)`, `pformat(%fmt)`, `sformat(%fmt)`, and `nolstretch`; see [R] [estimation options](#).

The following option is available with `xtivreg` but is not shown in the dialog box:

`coeflegend`; see [R] [estimation options](#).

Options for FE model

Model

`fe` requests the fixed-effects (within) regression estimator.

`regress` specifies that all the covariates are to be treated as exogenous and that the instrument list is to be ignored. Specifying `regress` causes `xtivreg` to fit the requested panel-data regression model of *depvar* on *varlist*₁ and *varlist*₂, ignoring *varlist*_{iv}.

SE

`vce(vcetype)` specifies the type of standard error reported, which includes types that are derived from asymptotic theory (`conventional`) and that use bootstrap or jackknife methods (`bootstrap`, `jackknife`); see [XT] [vce_options](#).

`vce(conventional)`, the default, uses the conventionally derived variance estimator for generalized least-squares regression.

Reporting

`level(#)`; see [R] [estimation options](#).

`first` specifies that the first-stage regressions be displayed.

`small` specifies that t statistics be reported instead of Z statistics and that F statistics be reported instead of χ^2 statistics.

display_options: `noomitted`, `vsquish`, `noemptycells`, `baselevels`, `allbaselevels`, `nofvlabel`, `fvwrap(#)`, `fvwrapon(style)`, `cformat(%fmt)`, `pformat(%fmt)`, `sformat(%fmt)`, and `nolstretch`; see [R] [estimation options](#).

The following option is available with `xtivreg` but is not shown in the dialog box:
`coeflegend`; see [R] [estimation options](#).

Options for FD model

Model

`noconstant`; see [R] [estimation options](#).

`fd` requests the first-differenced regression estimator.

`regress` specifies that all the covariates are to be treated as exogenous and that the instrument list is to be ignored. Specifying `regress` causes `xtivreg` to fit the requested panel-data regression model of *depvar* on *varlist*₁ and *varlist*₂, ignoring *varlist*_{iv}.

SE

`vce(vctype)` specifies the type of standard error reported, which includes types that are derived from asymptotic theory (`conventional`) and that use bootstrap or jackknife methods (`bootstrap`, `jackknife`); see [XT] [vce_options](#).

`vce(conventional)`, the default, uses the conventionally derived variance estimator for generalized least-squares regression.

Reporting

`level(#)`; see [R] [estimation options](#).

`first` specifies that the first-stage regressions be displayed.

`small` specifies that *t* statistics be reported instead of *Z* statistics and that *F* statistics be reported instead of χ^2 statistics.

display_options: `noomitted`, `vsquish`, `cformat(%fmt)`, `pformat(%fmt)`, `sformat(%fmt)`, and `no1stretch`; see [R] [estimation options](#).

The following option is available with `xtivreg` but is not shown in the dialog box:
`coeflegend`; see [R] [estimation options](#).

Remarks and examples

[stata.com](https://www.stata.com)

If you have not read [XT] [xt](#), please do so.

Consider an equation of the form

$$y_{it} = \mathbf{Y}_{it}\boldsymbol{\gamma} + \mathbf{X}_{1it}\boldsymbol{\beta} + \mu_i + \nu_{it} = \mathbf{Z}_{it}\boldsymbol{\delta} + \mu_i + \nu_{it} \quad (1)$$

where

y_{it} is the dependent variable;

$\mathbf{Y}_{it}$ is an $1 \times g_2$ vector of observations on g_2 endogenous variables included as covariates, and these variables are allowed to be correlated with the ν_{it} ;

$\mathbf{X}_{1it}$ is an $1 \times k_1$ vector of observations on the exogenous variables included as covariates;

$$\mathbf{Z}_{it} = [\mathbf{Y}_{it} \ \mathbf{X}_{it}];$$

γ is a $g_2 \times 1$ vector of coefficients;

β is a $k_1 \times 1$ vector of coefficients; and

δ is a $K \times 1$ vector of coefficients, where $K = g_2 + k_1$.

Assume that there is a $1 \times k_2$ vector of observations on the k_2 instruments in $\mathbf{X}_{2it}$. The order condition is satisfied if $k_2 \geq g_2$. Let $\mathbf{X}_{it} = [\mathbf{X}_{1it} \ \mathbf{X}_{2it}]$. `xtivreg` handles exogenously unbalanced panel data. Thus define T_i to be the number of observations on panel i , n to be the number of panels and N to be the total number of observations; that is, $N = \sum_{i=1}^n T_i$.

`xtivreg` offers five different estimators, which may be applied to models having the form of (1). The first-differenced estimator (FD2SLS) removes the μ_i by fitting the model in first differences. The within estimator (FE2SLS) fits the model after sweeping out the μ_i by removing the panel-level means from each variable. The between estimator (BE2SLS) models the panel averages. The two random-effects estimators, G2SLS and EC2SLS, treat the μ_i as random variables that are independent and identically distributed (i.i.d.) over the panels. Except for (FD2SLS), all of these estimators are generalizations of estimators in `xtreg`. See [XT] `xtreg` for a discussion of these estimators for exogenous covariates.

Although the estimators allow for different assumptions about the μ_i , all the estimators assume that the idiosyncratic error term ν_{it} has zero mean and is uncorrelated with the variables in $\mathbf{X}_{it}$. Just as when there are no endogenous covariates, as discussed in [XT] `xtreg`, there are various perspectives on what assumptions should be placed on the μ_i . If they are assumed to be fixed, the μ_i may be correlated with the variables in $\mathbf{X}_{it}$, and the within estimator is efficient within a class of limited information estimators. Alternatively, if the μ_i are assumed to be random, they are also assumed to be i.i.d. over the panels. If the μ_i are assumed to be uncorrelated with the variables in $\mathbf{X}_{it}$, the GLS random-effects estimators are more efficient than the within estimator. However, if the μ_i are correlated with the variables in $\mathbf{X}_{it}$, the random-effects estimators are inconsistent but the within estimator is consistent. The price of using the within estimator is that it is not possible to estimate coefficients on time-invariant variables, and all inference is conditional on the μ_i in the sample. See Mundlak (1978) and Hsiao (2003) for discussions of this interpretation of the within estimator.

► Example 1: Fixed-effects model

For the within estimator, consider another version of the wage equation discussed in [XT] `xtreg`. The data for this example come from an extract of women from the National Longitudinal Survey of Youth that was described in detail in [XT] `xt`. Restricting ourselves to only time-varying covariates, we might suppose that the log of the real wage was a function of the individual's age, age², her tenure in the observed place of employment, whether she belonged to union, whether she lives in metropolitan area, and whether she lives in the south. The variables for these are, respectively, `age`, `c.age#c.age`, `tenure`, `union`, `not_smsa`, and `south`. If we treat all the variables as exogenous, we can use the one-stage within estimator from `xtreg`, yielding

```
. use http://www.stata-press.com/data/r13/nlswork
(National Longitudinal Survey. Young Women 14-26 years of age in 1968)
. xtreg ln_w age c.age#c.age tenure not_smsa union south, fe

Fixed-effects (within) regression              Number of obs   =   19007
Group variable: idcode                       Number of groups =   4134

R-sq:  within = 0.1333                        Obs per group: min =    1
        between = 0.2375                      avg             =   4.6
        overall = 0.2031                      max             =   12

                                           F(6,14867)      =   381.19
corr(u_i, Xb) = 0.2074                      Prob > F         =   0.0000
```

ln_wage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
age	.0311984	.0033902	9.20	0.000	.0245533	.0378436
c.age#c.age	-.0003457	.0000543	-6.37	0.000	-.0004522	-.0002393
tenure	.0176205	.0008099	21.76	0.000	.0160331	.0192079
not_smsa	-.0972535	.0125377	-7.76	0.000	-.1218289	-.072678
union	.0975672	.0069844	13.97	0.000	.0838769	.1112576
south	-.0620932	.013327	-4.66	0.000	-.0882158	-.0359706
_cons	1.091612	.0523126	20.87	0.000	.9890729	1.194151
sigma_u	.3910683					
sigma_e	.25545969					
rho	.70091004	(fraction of variance due to u_i)				

F test that all u_i=0: F(4133, 14867) = 8.31 Prob > F = 0.0000

All the coefficients are statistically significant and have the expected signs.

Now suppose that we wish to model tenure as a function of union and south and that we believe that the errors in the two equations are correlated. Because we are still interested in the within estimates, we now need a two-stage least-squares estimator. The following output shows the command and the results from fitting this model:


```
. xtivreg ln_w age c.age#c.age not_smsa (tenure = union south), fe
Fixed-effects (within) IV regression      Number of obs      =      19007
Group variable: idcode                   Number of groups    =      4134
R-sq:  within = .
      between = 0.1304
      overall  = 0.0897
Obs per group: min =      1
              avg  =      4.6
              max  =      12
Wald chi2(4) =      147926.58
Prob > chi2  =      0.0000

corr(u_i, Xb) = -0.6843
```

ln_wage	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
tenure	.2403531	.0373419	6.44	0.000	.1671643	.3135419
age	.0118437	.0090032	1.32	0.188	-.0058023	.0294897
c.age#c.age	-.0012145	.0001968	-6.17	0.000	-.0016003	-.0008286
not_smsa	-.0167178	.0339236	-0.49	0.622	-.0832069	.0497713
_cons	1.678287	.1626657	10.32	0.000	1.359468	1.997106
sigma_u	.70661941	(fraction of variance due to u_i)				
sigma_e	.63029359					
rho	.55690561					
F test that all u_i=0:		F(4133,14869) =		1.44	Prob > F	= 0.0000
Instrumented:	tenure					
Instruments:	age c.age#c.age not_smsa union south					

Although all the coefficients still have the expected signs, the coefficients on `age` and `not_smsa` are no longer statistically significant. Given that these variables have been found to be important in many other studies, we might want to rethink our specification. ◀

If we are willing to assume that the μ_i are uncorrelated with the other covariates, we can fit a random-effects model. The model is frequently known as the variance-components or error-components model. `xtivreg` has estimators for two-stage least-squares one-way error-components models. In the one-way framework, there are two variance components to estimate, the variance of the μ_i and the variance of the ν_{it} . Because the variance components are unknown, consistent estimates are required to implement feasible GLS. `xtivreg` offers two choices: a Swamy–Arora method and simple consistent estimators from Baltagi and Chang (2000).

Baltagi and Chang (1994) derived the Swamy–Arora estimators of the variance components for unbalanced panels. By default, `xtivreg` uses estimators that extend these unbalanced Swamy–Arora estimators to the case with instrumental variables. The default Swamy–Arora method contains a degree-of-freedom correction to improve its performance in small samples. Baltagi and Chang (2000) use variance-components estimators, which are based on the ideas of Amemiya (1971) and Swamy and Arora (1972), but they do not attempt to make small-sample adjustments. These consistent estimators of the variance components will be used if the `nosa` option is specified.

Using either estimator of the variance components, `xtivreg` offers two GLS estimators of the random-effects model. These two estimators differ only in how they construct the GLS instruments from the exogenous and instrumental variables contained in $\mathbf{X}_{it} = [\mathbf{X}_{1it} \ \mathbf{X}_{2it}]$. The default method, G2SLS, which is from Balestra and Varadharajan-Krishnakumar, uses the exogenous variables after they have been passed through the feasible GLS transform. In math, G2SLS uses $\mathbf{X}_{it}^*$ for the GLS instruments, where $\mathbf{X}_{it}^*$ is constructed by passing each variable in $\mathbf{X}_{it}$ through the GLS transform in (3) given in *Methods and formulas*. If the `ec2sls` option is specified, `xtivreg` performs Baltagi’s

EC2SLS. In EC2SLS, the instruments are $\tilde{\mathbf{X}}_{it}$ and $\overline{\mathbf{X}}_{it}$, where $\tilde{\mathbf{X}}_{it}$ is constructed by passing each of the variables in $\mathbf{X}_{it}$ through the within transform, and $\overline{\mathbf{X}}_{it}$ is constructed by passing each variable through the between transform. The within and between transforms are given in the [Methods and formulas](#) section. Baltagi and Li (1992) show that, although the G2SLS instruments are a subset of those contained in EC2SLS, the extra instruments in EC2SLS are redundant in the sense of White (2001). Given the extra computational cost, G2SLS is the default.

► Example 2: GLS random-effects model

Here is the output from applying the G2SLS estimator to this model:

```
. xtivreg ln_w age c.age#c.age not_smsa 2.race (tenure = union birth south), re
G2SLS random-effects IV regression                Number of obs      =    19007
Group variable: idcode                          Number of groups   =     4134
R-sq:  within  = 0.0664                        Obs per group: min =         1
        between = 0.2098                        avg             =        4.6
        overall = 0.1463                        max             =        12

corr(u_i, X)      = 0 (assumed)                  Wald chi2(5)       =   1446.37
                                                Prob > chi2        =    0.0000
```

ln_wage	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
tenure	.1391798	.0078756	17.67	0.000	.123744	.1546157
age	.0279649	.0054182	5.16	0.000	.0173454	.0385843
c.age#c.age	-.0008357	.0000871	-9.60	0.000	-.0010063	-.000665
not_smsa	-.2235103	.0111371	-20.07	0.000	-.2453386	-.2016821
race						
black	-.2078613	.0125803	-16.52	0.000	-.2325183	-.1832044
_cons	1.337684	.0844988	15.83	0.000	1.172069	1.503299
sigma_u	.36582493					
sigma_e	.63031479					
rho	.25197078	(fraction of variance due to u_i)				

```
Instrumented:  tenure
Instruments:   age c.age#c.age not_smsa 2.race union birth_yr south
```

We have included two time-invariant covariates, `birth_yr` and `2.race`. All the coefficients are statistically significant and are of the expected sign.

Applying the EC2SLS estimator yields similar results:

```
. xtivreg ln_w age c.age#c.age not_smsa 2.race (tenure = union birth south), re
> ec2sls
```

EC2SLS random-effects IV regression	Number of obs	=	19007		
Group variable: idcode	Number of groups	=	4134		
R-sq: within	=	0.0898	Obs per group: min	=	1
between	=	0.2608	avg	=	4.6
overall	=	0.1926	max	=	12
corr(u_i, X)	=	0 (assumed)	Wald chi2(5)	=	2721.92
			Prob > chi2	=	0.0000

ln_wage	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
tenure	.064822	.0025647	25.27	0.000	.0597953	.0698486
age	.0380048	.0039549	9.61	0.000	.0302534	.0457562
c.age#c.age	-.0006676	.0000632	-10.56	0.000	-.0007915	-.0005438
not_smsa	-.2298961	.0082993	-27.70	0.000	-.2461625	-.2136297
race						
black	-.1823627	.0092005	-19.82	0.000	-.2003954	-.16433
_cons	1.110564	.0606538	18.31	0.000	.9916849	1.229443
sigma_u	.36582493					
sigma_e	.63031479					
rho	.25197078	(fraction of variance due to u_i)				

Instrumented: tenure
Instruments: age c.age#c.age not_smsa 2.race union birth_yr south

Fitting the same model as above with the G2SLS estimator and the consistent variance components estimators yields

```
. xtivreg ln_w age c.age#c.age not_smsa 2.race (tenure = union birth south), re
> nosa

G2SLS random-effects IV regression               Number of obs      =    19007
Group variable: idcode                          Number of groups   =    4134

R-sq:  within = 0.0664                          Obs per group: min =         1
        between = 0.2098                        avg               =         4.6
        overall = 0.1463                        max               =        12

                                         Wald chi2(5)       =    1446.93
                                         Prob > chi2        =     0.0000

corr(u_i, X)      = 0 (assumed)
```

ln_wage	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
tenure	.1391859	.007873	17.68	0.000	.1237552	.1546166
age	.0279697	.005419	5.16	0.000	.0173486	.0385909
c.age#c.age	-.0008357	.0000871	-9.60	0.000	-.0010064	-.000665
not_smsa	-.2235738	.0111344	-20.08	0.000	-.2453967	-.2017508
race						
black	-.2078733	.0125751	-16.53	0.000	-.2325201	-.1832265
_cons	1.337522	.0845083	15.83	0.000	1.171889	1.503155
sigma_u	.36535633					
sigma_e	.63020883					
rho	.2515512	(fraction of variance due to u_i)				
Instrumented:	tenure					
Instruments:	age c.age#c.age not_smsa 2.race union birth_yr south					

◀

➤ **Example 3: First-differenced estimator**

The two-stage least-squares first-differenced estimator (FD2SLS) has been used to fit both fixed-effect and random-effect models. If the μ_i are truly fixed-effects, the FD2SLS estimator is not as efficient as the two-stage least-squares within estimator for finite T_i . Similarly, if none of the endogenous variables are lagged dependent variables, the exogenous variables are all strictly exogenous, and the random effects are i.i.d. and independent of the $\mathbf{X}_{it}$, the two-stage GLS estimators are more efficient than the FD2SLS estimator. However, the FD2SLS estimator has been used to obtain consistent estimates when one of these conditions fails. [Anderson and Hsiao \(1981\)](#) used a version of the FD2SLS estimator to fit a panel-data model with a lagged dependent variable.

[Arellano and Bond \(1991\)](#) develop new one-step and two-step GMM estimators for dynamic panel data. See [\[XT\] xtabond](#) for a discussion of these estimators and Stata’s implementation of them. In their article, [Arellano and Bond \(1991\)](#) apply their new estimators to a model of dynamic labor demand that had previously been considered by [Layard and Nickell \(1986\)](#). They also compare the results of their estimators with those from the Anderson–Hsiao estimator using data from an unbalanced panel of firms from the United Kingdom. As is conventional, all variables are indexed over the firm i and time t . In this dataset, n_{it} is the log of employment in firm i inside the United Kingdom at time t , w_{it} is the natural log of the real product wage, k_{it} is the natural log of the gross capital stock, and ys_{it} is the natural log of industry output. The model also includes time dummies `yr1980`, `yr1981`, `yr1982`, `yr1983`, and `yr1984`. In [Arellano and Bond \(1991, table 5, column e\)](#), the authors present the results from applying one version of the Anderson–Hsiao estimator to these data. This example reproduces their results for the coefficients, though standard errors are different because Arellano and Bond are using robust standard errors.

```

. use http://www.stata-press.com/data/r13/abdata
. xtivreg n l2.n l(0/1).w l(0/2).(k ys) yr1981-yr1984 (l.n = l3.n), fd
First-differenced IV regression
Group variable:   id                Number of obs   =      471
Time variable:   year              Number of groups =      140
R-sq:  within    = 0.0141           Obs per group: min =       3
       between    = 0.9165                             avg   =      3.4
       overall    = 0.9892                             max   =       5
                                                    Wald chi2(14)    =     122.53
corr(u_i, Xb)    = 0.9239              Prob > chi2       =      0.0000

```

D.n	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
n						
LD.	1.422765	1.583053	0.90	0.369	-1.679962	4.525493
L2D.	-.1645517	.1647179	-1.00	0.318	-.4873928	.1582894
w						
D1.	-.7524675	.1765733	-4.26	0.000	-1.098545	-.4063902
LD.	.9627611	1.086506	0.89	0.376	-1.166752	3.092275
k						
D1.	.3221686	.1466086	2.20	0.028	.0348211	.6095161
LD.	-.3248778	.5800599	-0.56	0.575	-1.461774	.8120187
L2D.	-.0953947	.1960883	-0.49	0.627	-.4797207	.2889314
ys						
D1.	.7660906	.369694	2.07	0.038	.0415037	1.490678
LD.	-1.361881	1.156835	-1.18	0.239	-3.629237	.9054744
L2D.	.3212993	.5440403	0.59	0.555	-.745	1.387599
yr1981						
D1.	-.0574197	.0430158	-1.33	0.182	-.1417291	.0268896
yr1982						
D1.	-.0882952	.0706214	-1.25	0.211	-.2267106	.0501203
yr1983						
D1.	-.1063153	.10861	-0.98	0.328	-.319187	.1065563
yr1984						
D1.	-.1172108	.15196	-0.77	0.441	-.4150468	.1806253
_cons	.0161204	.0336264	0.48	0.632	-.0497861	.082027
sigma_u	.29069213					
sigma_e	.18855982					
rho	.70384993	(fraction of variance due to u_i)				

```

Instrumented:  L.n
Instruments:   L2.n w L.w k L.k L2.k ys L.ys L2.ys yr1981 yr1982 yr1983
                yr1984 L3.n

```

Stored results

`xtivreg`, `re` stores the following in `e()`:

Scalars

<code>e(N)</code>	number of observations
<code>e(N_g)</code>	number of groups
<code>e(df_m)</code>	model degrees of freedom
<code>e(df_rz)</code>	residual degrees of freedom
<code>e(g_min)</code>	smallest group size
<code>e(g_avg)</code>	average group size
<code>e(g_max)</code>	largest group size
<code>e(Tcon)</code>	1 if panels balanced; 0 otherwise
<code>e(sigma)</code>	ancillary parameter (γ , <code>lnormal</code>)
<code>e(sigma_u)</code>	panel-level standard deviation
<code>e(sigma_e)</code>	standard deviation of ϵ_{it}
<code>e(r2_w)</code>	R -squared for within model
<code>e(r2_o)</code>	R -squared for overall model
<code>e(r2_b)</code>	R -squared for between model
<code>e(chi2)</code>	χ^2
<code>e(rho)</code>	ρ
<code>e(F)</code>	model F (small only)
<code>e(m_p)</code>	p -value from model test
<code>e(thta_min)</code>	minimum θ
<code>e(thta_5)</code>	θ , 5th percentile
<code>e(thta_50)</code>	θ , 50th percentile
<code>e(thta_95)</code>	θ , 95th percentile
<code>e(thta_max)</code>	maximum θ
<code>e(Tbar)</code>	harmonic mean of group sizes
<code>e(rank)</code>	rank of <code>e(V)</code>

Macros

<code>e(cmd)</code>	<code>xtivreg</code>
<code>e(cmdline)</code>	command as typed
<code>e(depvar)</code>	name of dependent variable
<code>e(ivar)</code>	variable denoting groups
<code>e(tvar)</code>	variable denoting time within groups
<code>e(insts)</code>	instruments
<code>e(instd)</code>	instrumented variables
<code>e(model)</code>	<code>g2s1s</code> or <code>ec2s1s</code>
<code>e(small)</code>	small, if specified
<code>e(chi2type)</code>	Wald; type of model χ^2 test
<code>e(vce)</code>	<code>vcetype</code> specified in <code>vce()</code>
<code>e(vcetype)</code>	title used to label Std. Err.
<code>e(properties)</code>	<code>b V</code>
<code>e(predict)</code>	program used to implement <code>predict</code>
<code>e(marginsok)</code>	predictions allowed by <code>margins</code>
<code>e(marginsnotok)</code>	predictions disallowed by <code>margins</code>
<code>e(asbalanced)</code>	factor variables <code>fvset</code> as <code>asbalanced</code>
<code>e(asobserved)</code>	factor variables <code>fvset</code> as <code>asobserved</code>

Matrices

<code>e(b)</code>	coefficient vector
<code>e(V)</code>	variance–covariance matrix of the estimators

Functions

<code>e(sample)</code>	marks estimation sample
------------------------	-------------------------

xtivreg, **be** stores the following in `e()`:

Scalars

<code>e(N)</code>	number of observations
<code>e(N_g)</code>	number of groups
<code>e(mss)</code>	model sum of squares
<code>e(df_m)</code>	model degrees of freedom
<code>e(rss)</code>	residual sum of squares
<code>e(df_r)</code>	residual degrees of freedom
<code>e(df_rz)</code>	residual degrees of freedom for the between-transformed regression
<code>e(g_min)</code>	smallest group size
<code>e(g_avg)</code>	average group size
<code>e(g_max)</code>	largest group size
<code>e(rs_a)</code>	adjusted R^2
<code>e(r2_w)</code>	R -squared for within model
<code>e(r2_o)</code>	R -squared for overall model
<code>e(r2_b)</code>	R -squared for between model
<code>e(chi2)</code>	model Wald
<code>e(chi2_p)</code>	p -value for model χ^2 test
<code>e(F)</code>	F statistic (small only)
<code>e(rmse)</code>	root mean squared error
<code>e(rank)</code>	rank of <code>e(V)</code>

Macros

<code>e(cmd)</code>	xtivreg
<code>e(cmdline)</code>	command as typed
<code>e(depvar)</code>	name of dependent variable
<code>e(ivar)</code>	variable denoting groups
<code>e(tvar)</code>	variable denoting time within groups
<code>e(insts)</code>	instruments
<code>e(instd)</code>	instrumented variables
<code>e(model)</code>	be
<code>e(small)</code>	small , if specified
<code>e(vce)</code>	<i>vcetype</i> specified in <code>vce()</code>
<code>e(vcetype)</code>	title used to label Std. Err.
<code>e(properties)</code>	b V
<code>e(predict)</code>	program used to implement predict
<code>e(marginsok)</code>	predictions allowed by margins
<code>e(marginsnotok)</code>	predictions disallowed by margins
<code>e(asbalanced)</code>	factor variables <i>fvset</i> as asbalanced
<code>e(asobserved)</code>	factor variables <i>fvset</i> as asobserved

Matrices

<code>e(b)</code>	coefficient vector
<code>e(V)</code>	variance–covariance matrix of the estimators

Functions

<code>e(sample)</code>	marks estimation sample
------------------------	-------------------------

xtivreg, **fe** stores the following in **e()**:

Scalars

e(N)	number of observations
e(N_g)	number of groups
e(df_m)	model degrees of freedom
e(rss)	residual sum of squares
e(df_r)	residual degrees of freedom (small only)
e(df_rz)	residual degrees of freedom for the within-transformed regression
e(g_min)	smallest group size
e(g_avg)	average group size
e(g_max)	largest group size
e(sigma)	ancillary parameter (γ , $\lnormal$)
e(corr)	$\text{corr}(u_i, Xb)$
e(sigma_u)	panel-level standard deviation
e(sigma_e)	standard deviation of ϵ_{it}
e(r2_w)	R -squared for within model
e(r2_o)	R -squared for overall model
e(r2_b)	R -squared for between model
e(chi2)	model Wald (not small)
e(df_b)	degrees of freedom for χ^2 statistic
e(chi2_p)	p -value for model χ^2 statistic
e(rho)	ρ
e(F)	F statistic (small only)
e(F_f)	F for $H_0: u_i=0$
e(F_fp)	p -value for F for $H_0: u_i=0$
e(df_a)	degrees of freedom for absorbed effect
e(rank)	rank of e(V)

Macros

e(cmd)	xtivreg
e(cmdline)	command as typed
e(depvar)	name of dependent variable
e(ivar)	variable denoting groups
e(tvar)	variable denoting time within groups
e(insts)	instruments
e(instd)	instrumented variables
e(model)	fe
e(small)	small, if specified
e(vce)	<i>vcetype</i> specified in vce()
e(vcetype)	title used to label Std. Err.
e(properties)	b V
e(predict)	program used to implement predict
e(marginsok)	predictions allowed by margins
e(marginsnotok)	predictions disallowed by margins
e(asbalanced)	factor variables fvset as asbalanced
e(asobserved)	factor variables fvset as asobserved

Matrices

e(b)	coefficient vector
e(V)	variance–covariance matrix of the estimators

Functions

e(sample)	marks estimation sample
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`xtivreg, fd` stores the following in `e()`:

Scalars

<code>e(N)</code>	number of observations
<code>e(N_g)</code>	number of groups
<code>e(rss)</code>	residual sum of squares
<code>e(df_r)</code>	residual degrees of freedom (small only)
<code>e(df_rz)</code>	residual degrees of freedom for first-differenced regression
<code>e(g_min)</code>	smallest group size
<code>e(g_avg)</code>	average group size
<code>e(g_max)</code>	largest group size
<code>e(sigma)</code>	ancillary parameter (γ , <code>lnormal</code>)
<code>e(corr)</code>	$\text{corr}(u_i, Xb)$
<code>e(sigma_u)</code>	panel-level standard deviation
<code>e(sigma_e)</code>	standard deviation of ϵ_{it}
<code>e(r2_w)</code>	R -squared for within model
<code>e(r2_o)</code>	R -squared for overall model
<code>e(r2_b)</code>	R -squared for between model
<code>e(chi2)</code>	model Wald (not small)
<code>e(df_b)</code>	degrees of freedom for the χ^2 statistic
<code>e(chi2_p)</code>	p -value for model χ^2 statistic
<code>e(rho)</code>	ρ
<code>e(F)</code>	F statistic (small only)
<code>e(rank)</code>	rank of <code>e(V)</code>

Macros

<code>e(cmd)</code>	<code>xtivreg</code>
<code>e(cmdline)</code>	command as typed
<code>e(depvar)</code>	name of dependent variable
<code>e(ivar)</code>	variable denoting groups
<code>e(tvar)</code>	variable denoting time within groups
<code>e(insts)</code>	instruments
<code>e(instd)</code>	instrumented variables
<code>e(model)</code>	<code>fd</code>
<code>e(small)</code>	small, if specified
<code>e(vce)</code>	<code>vcetype</code> specified in <code>vce()</code>
<code>e(vcetype)</code>	title used to label Std. Err.
<code>e(properties)</code>	<code>b V</code>
<code>e(predict)</code>	program used to implement <code>predict</code>
<code>e(marginsok)</code>	predictions allowed by <code>margins</code>

Matrices

<code>e(b)</code>	coefficient vector
<code>e(V)</code>	variance–covariance matrix of the estimators

Functions

<code>e(sample)</code>	marks estimation sample
------------------------	-------------------------

Methods and formulas

Consider an equation of the form

$$y_{it} = \mathbf{Y}_{it}\boldsymbol{\gamma} + \mathbf{X}_{1it}\boldsymbol{\beta} + \mu_i + \nu_{it} = \mathbf{Z}_{it}\boldsymbol{\delta} + \mu_i + \nu_{it} \quad (2)$$

where

y_{it} is the dependent variable;

$\mathbf{Y}_{it}$ is an $1 \times g_2$ vector of observations on g_2 endogenous variables included as covariates, and these variables are allowed to be correlated with the ν_{it} ;

$\mathbf{X}_{1it}$ is an $1 \times k_1$ vector of observations on the exogenous variables included as covariates;

$\mathbf{Z}_{it} = [\mathbf{Y}_{it} \ \mathbf{X}_{1it}]$;

$\boldsymbol{\gamma}$ is a $g_2 \times 1$ vector of coefficients;

$\boldsymbol{\beta}$ is a $k_1 \times 1$ vector of coefficients; and

$\boldsymbol{\delta}$ is a $K \times 1$ vector of coefficients, where $K = g_2 + k_1$.

Assume that there is a $1 \times k_2$ vector of observations on the k_2 instruments in $\mathbf{X}_{2it}$. The order condition is satisfied if $k_2 \geq g_2$. Let $\mathbf{X}_{it} = [\mathbf{X}_{1it} \ \mathbf{X}_{2it}]$. **xtivreg** handles exogenously unbalanced panel data. Thus define T_i to be the number of observations on panel i , n to be the number of panels, and N to be the total number of observations; that is, $N = \sum_{i=1}^n T_i$.

Methods and formulas are presented under the following headings:

xtivreg, fd
xtivreg, fe
xtivreg, be
xtivreg, re

xtivreg, fd

As the name implies, this estimator obtains its estimates and conventional VCE from an instrumental-variables regression on the first-differenced data. Specifically, first differencing the data yields

$$y_{it} - y_{it-1} = (\mathbf{Z}_{it} - \mathbf{Z}_{i,t-1})\boldsymbol{\delta} + \nu_{it} - \nu_{i,t-1}$$

With the μ_i removed by differencing, we can obtain the estimated coefficients and their estimated variance–covariance matrix from a standard two-stage least-squares regression of Δy_{it} on $\Delta \mathbf{Z}_{it}$ with instruments $\Delta \mathbf{X}_{it}$.

R^2 within is reported as $\left[\text{corr}\{(\mathbf{Z}_{it} - \bar{\mathbf{Z}}_i)\hat{\boldsymbol{\delta}}, y_{it} - \bar{y}_i\} \right]^2$.

R^2 between is reported as $\left\{ \text{corr}(\bar{\mathbf{Z}}_i\hat{\boldsymbol{\delta}}, \bar{y}_i) \right\}^2$.

R^2 overall is reported as $\left\{ \text{corr}(\mathbf{Z}_{it}\hat{\boldsymbol{\delta}}, y_{it}) \right\}^2$.

xtivreg, fe

At the heart of this model is the within transformation. The within transform of a variable w is

$$\tilde{w}_{it} = w_{it} - \bar{w}_i + \bar{w}$$

where

$$\bar{w}_i = \frac{1}{n} \sum_{t=1}^{T_i} w_{it}$$

$$\bar{w} = \frac{1}{N} \sum_{i=1}^n \sum_{t=1}^{T_i} w_{it}$$

and n is the number of groups and N is the total number of observations on the variable.

The within transform of (2) is

$$\tilde{y}_{it} = \tilde{\mathbf{Z}}_{it} + \tilde{\nu}_{it}$$

The within transform has removed the μ_i . With the μ_i gone, the within 2SLS estimator can be obtained from a two-stage least-squares regression of $\tilde{y}_{it}$ on $\tilde{\mathbf{Z}}_{it}$ with instruments $\tilde{\mathbf{X}}_{it}$.

Suppose that there are K variables in $\mathbf{Z}_{it}$, including the mandatory constant. There are $K + n - 1$ parameters estimated in the model, and the conventional VCE for the within estimator is

$$\frac{N - K}{N - n - K + 1} V_{IV}$$

where V_{IV} is the VCE from the above two-stage least-squares regression.

From the estimate of $\hat{\boldsymbol{\delta}}$, estimates $\hat{\mu}_i$ of μ_i are obtained as $\hat{\mu}_i = \bar{y}_i - \bar{\mathbf{Z}}_i \hat{\boldsymbol{\delta}}$. Reported from the calculated $\hat{\mu}_i$ is its standard deviation and its correlation with $\bar{\mathbf{Z}}_i \hat{\boldsymbol{\delta}}$. Reported as the standard deviation of ν_{it} is the regression's estimated root mean squared error, s^2 , which is adjusted (as previously stated) for the $n - 1$ estimated means.

R^2 within is reported as the R^2 from the mean-deviated regression.

R^2 between is reported as $\left\{ \text{corr}(\bar{\mathbf{Z}}_i \hat{\boldsymbol{\delta}}, \bar{y}_i) \right\}^2$.

R^2 overall is reported as $\left\{ \text{corr}(\mathbf{Z}_{it} \hat{\boldsymbol{\delta}}, y_{it}) \right\}^2$.

At the bottom of the output, an F statistics against the null hypothesis that all the μ_i are zero is reported. This F statistic is an application of the results in [Wooldridge \(1990\)](#).

xtivreg, be

After passing (2) through the between transform, we are left with

$$\bar{y}_i = \alpha + \bar{\mathbf{Z}}_i \boldsymbol{\delta} + \mu_i + \bar{\nu}_i \quad (3)$$

where

$$\bar{w}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} w_{it} \quad \text{for } w \in \{y, \mathbf{Z}, \nu\}$$

Similarly, define $\bar{\mathbf{X}}_i$ as the matrix of instruments $\mathbf{X}_{it}$ after they have been passed through the between transform.

The BE2SLS estimator of (3) obtains its coefficient estimates and its conventional VCE, a two-stage least-squares regression of $\bar{y}_i$ on $\bar{Z}_i$ with instruments $\bar{X}_i$ in which each average appears T_i times.

R^2 between is reported as the R^2 from the fitted regression.

R^2 within is reported as $\left[\text{corr}\{(\mathbf{Z}_{it} - \bar{\mathbf{Z}}_i)\hat{\boldsymbol{\delta}}, y_{it} - \bar{y}_i\} \right]^2$.

R^2 overall is reported as $\left\{ \text{corr}(\mathbf{Z}_{it}\hat{\boldsymbol{\delta}}, y_{it}) \right\}^2$.

xtivreg, re

Per Baltagi and Chang (2000), let

$$u = \mu_i + \nu_{it}$$

be the $N \times 1$ vector of combined errors. Then under the assumptions of the random-effects model,

$$E(uu') = \sigma_\nu^2 \text{diag} \left[I_{T_i} - \frac{1}{T_i} \boldsymbol{\nu}_{T_i} \boldsymbol{\nu}_{T_i}' \right] + \text{diag} \left[w_i \frac{1}{T_i} \boldsymbol{\nu}_{T_i} \boldsymbol{\nu}_{T_i}' \right]$$

where

$$\omega_i = T_i \sigma_\mu^2 + \sigma_\nu^2$$

and $\boldsymbol{\nu}_{T_i}$ is a vector of ones of dimension T_i .

Because the variance components are unknown, consistent estimates are required to implement feasible GLS. `xtivreg` offers two choices. The default is a simple extension of the Swamy–Arora method for unbalanced panels.

Let

$$u_{it}^w = \tilde{y}_{it} - \tilde{\mathbf{Z}}_{it} \hat{\boldsymbol{\delta}}_w$$

be the combined residuals from the within estimator. Let $\tilde{u}_{it}$ be the within-transformed u_{it} . Then

$$\hat{\sigma}_\nu^2 = \frac{\sum_{i=1}^n \sum_{t=1}^{T_i} \tilde{u}_{it}^2}{N - n - K + 1}$$

Let

$$u_{it}^b = y_{it} - \mathbf{Z}_{it} \boldsymbol{\delta}_b$$

be the combined residual from the between estimator. Let $\bar{u}_{it}^b$ be the between residuals after they have been passed through the between transform. Then

$$\hat{\sigma}_\mu^2 = \frac{\sum_{i=1}^n \sum_{t=1}^{T_i} \bar{u}_{it}^2 - (n - K) \hat{\sigma}_\nu^2}{N - r}$$

where

$$r = \text{trace} \left\{ \left(\bar{\mathbf{Z}}_i' \bar{\mathbf{Z}}_i \right)^{-1} \bar{\mathbf{Z}}_i' \mathbf{Z}_\mu \mathbf{Z}_\mu' \bar{\mathbf{Z}}_i \right\}$$

where

$$\mathbf{Z}_\mu = \text{diag} \left(\boldsymbol{\nu}_{T_i} \boldsymbol{\nu}_{T_i}' \right)$$

If the `nosa` option is specified, the consistent estimators described in Baltagi and Chang (2000) are used. These are given by

$$\hat{\sigma}_\nu^2 = \frac{\sum_{i=1}^n \sum_{t=1}^{T_i} \tilde{u}_{it}^2}{N - n}$$

and

$$\hat{\sigma}_\mu^2 = \frac{\sum_{i=1}^n \sum_{t=1}^{T_i} \bar{u}_{it}^2 - n\hat{\sigma}_\nu^2}{N}$$

The default Swamy–Arora method contains a degree-of-freedom correction to improve its performance in small samples.

Given estimates of the variance components, $\hat{\sigma}_\nu^2$ and $\hat{\sigma}_\mu^2$, the feasible GLS transform of a variable w is

$$w^* = w_{it} - \hat{\theta}_{it} \bar{w}_i. \quad (4)$$

where

$$\bar{w}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} w_{it}$$

$$\hat{\theta}_{it} = 1 - \left(\frac{\hat{\sigma}_\nu^2}{\hat{\omega}_i} \right)^{-\frac{1}{2}}$$

and

$$\hat{\omega}_i = T_i \hat{\sigma}_\mu^2 + \hat{\sigma}_\nu^2$$

Using either estimator of the variance components, `xtivreg` contains two GLS estimators of the random-effects model. These two estimators differ only in how they construct the GLS instruments from the exogenous and instrumental variables contained in $\mathbf{X}_{it} = [\mathbf{X}_{1it} \mathbf{X}_{2it}]$. The default method, G2SLS, which is from Balestra and Varadharajan-Krishnakumar, uses the exogenous variables after they have been passed through the feasible GLS transform. Mathematically, G2SLS uses $\mathbf{X}^*$ for the GLS instruments, where $\mathbf{X}^*$ is constructed by passing each variable in $\mathbf{X}$ through the GLS transform in (4). The G2SLS estimator obtains its coefficient estimates and conventional VCE from an instrumental variable regression of y_{it}^* on $\mathbf{Z}_{it}^*$ with instruments $\mathbf{X}_{it}^*$.

If the `ec2sls` option is specified, `xtivreg` performs Baltagi's EC2SLS. In EC2SLS, the instruments are $\tilde{\mathbf{X}}_{it}$ and $\bar{\mathbf{X}}_{it}$, where $\tilde{\mathbf{X}}_{it}$ is constructed by each of the variables in $\mathbf{X}_{it}$ throughout the GLS transform in (4), and $\bar{\mathbf{X}}_{it}$ is made of the group means of each variable in $\mathbf{X}_{it}$. The EC2SLS estimator obtains its coefficient estimates and its VCE from an instrumental variables regression of y_{it}^* on $\mathbf{Z}_{it}^*$ with instruments $\tilde{\mathbf{X}}_{it}$ and $\bar{\mathbf{X}}_{it}$.

Baltagi and Li (1992) show that although the G2SLS instruments are a subset of those in EC2SLS, the extra instruments in EC2SLS are redundant in the sense of White (2001). Given the extra computational cost, G2SLS is the default.

The standard deviation of $\mu_i + \nu_{it}$ is calculated as $\sqrt{\hat{\sigma}_\mu^2 + \hat{\sigma}_\nu^2}$.

R^2 between is reported as $\left\{ \text{corr}(\overline{\mathbf{Z}}_i \widehat{\boldsymbol{\delta}}, \overline{y}_i) \right\}^2$.

R^2 within is reported as $\left[\text{corr}\{(\mathbf{Z}_{it} - \overline{\mathbf{Z}}_i) \widehat{\boldsymbol{\delta}}, y_{it} - \overline{y}_i\} \right]^2$.

R^2 overall is reported as $\left\{ \text{corr}(\mathbf{Z}_{it} \widehat{\boldsymbol{\delta}}, y_{it}) \right\}^2$.

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References

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Also see

- [XT] **xtivreg postestimation** — Postestimation tools for `xtivreg`
- [XT] **xtset** — Declare data to be panel data
- [XT] **xtabond** — Arellano–Bond linear dynamic panel-data estimation
- [XT] **xthtaylor** — Hausman–Taylor estimator for error-components models
- [XT] **xtreg** — Fixed-, between-, and random-effects and population-averaged linear models
- [R] **ivregress** — Single-equation instrumental-variables regression
- [U] **20 Estimation and postestimation commands**